Research — Heroic Games Launcher

Nov 4, 2026 · @Tim

Summary

Heroic is our main import source for Epic, GOG, Amazon and ZOOM games, and an easy one: its library, playtime, favourites and categories are plain JSON under `~/.config/heroic`, and any game launches through `heroic://launch` with a no-GUI flag. It is a GPL-3.0 Electron + React + TypeScript launcher (stable v2.22.4, 2026-10-15) that drives a CLI backend per store: legendary, gogdl, nile and its own ZOOM code. Its web-styled UI pairs a collapsible sidebar with a 3:4 cover grid, a filter dropdown, a tabbed game page and a very large settings model, with umu as the default for Proton. Since v2.21 (April 2026) it also ships a fullscreen Console Mode for controllers.

We should read Heroic rather than replace it: import its games, show its playtime and art, and hand launches, installs and per-game Wine settings back to Heroic.

Project facts

Heroic is the most-installed reference app here (about 88k Flathub downloads a month) and releases roughly monthly.

Field	Value
Repo	Heroic-Games-Launcher/HeroicGamesLauncher
Latest release	v2.22.4, 2026-10-15
Recent releases	v2.22.0 (2026-05-16), v2.21.0 (2026-04-22, Console Mode, GOG achievements), v2.20.0 (2026-02-20), v2.19.0 (2026-01-27, ZOOM Platform, AppImage auto-update)
Licence	GPL-3.0-only
Stack	Electron 43, React 18.3, TypeScript 5.8, MUI 5, electron-vite, zustand, i18next (about 40 languages via Weblate), fuse.js fuzzy search

Field	Value
Store backends	`legendary` (Epic), `gog` (gogdl, plus comet for Galaxy features), `nile` (Amazon), `zoom` (ZOOM Platform, experimental, off by default), `sideload` (any app added by hand)
Third-party titles	EA app and Ubisoft Connect games bought on Epic are flagged as third-party managed
Compatibility	Wine, Proton, umu (default for Proton), Steam Runtime via umu for native games; Crossover and Game Porting Toolkit on macOS
Distribution	Flathub `com.heroicgameslauncher.hgl`, AppImage, deb, rpm, pacman, tar.xz, Windows, macOS; AUR `heroic-games-launcher-bin`; nixpkgs, WinGet, Homebrew

Library UI and navigation

Heroic's library is a sidebar plus a header of search, categories and filters over a 3:4 grid, with rich per-card status but alphabetical-only sorting.

Sidebar: Library, Stores (Epic, GOG, Amazon, ZOOM when enabled), Deals, Settings (General, Game Defaults, Advanced, System Info, Log), Console Mode, Downloads, Wine Manager, Manage Accounts, Accessibility, Documentation, then donation links. It is resizable, collapses below 120 px, and shows the current download and a Quit button.

Header

- Fuzzy search (Fuse, threshold 0.4, also matches normalised titles).
- Categories dropdown: custom categories plus Uncategorized, Select All, Manage Categories.
- Filters dropdown with store toggles (Epic, GOG, Amazon, ZOOM, Other), platform toggles (Windows, Linux or Mac, Browser), and view toggles (show hidden, show unavailable, favourites only, installed only, offline-capable only, third-party managed only, has updates only).

Library header row: "All Games" or "Favourites" with a count and Add Game, then icons for list or grid layout, sort ascending or descending (title only, ignoring a leading "The"), sort by status (installed, installing, not installed; on by default), an A–Z/# alphabet filter,

and Refresh. No sort by playtime, last played or size exists.

Top lane: a setting picks Disabled (default), Recently played, Recently played installed only, or Favourites; the first card in that lane uses a wider 16:10 `art_cover` variant.

Cards

- Grid columns `repeat(auto-fill, minmax(156px, 1fr))`; card aspect 173:275 with a portrait 3:4 `art_square` image, optional centred logo, title and store below.
- Not-installed games are grayscale; during install the grayscale fades with progress and a status label appears top left. Accessibility settings can force colour and always-visible titles.
- Hover shows update, settings and play, stop, install or cancel buttons.
- Statuses: installing, importing, updating, launching, playing, uninstalling, repairing, moving, queued, syncing saves, redist, extracting, winetricks, not available, not supported.
- The list layout uses columns of 3fr, 2fr, 1fr, 10%.

Context menu (depends on state): Remove from Queue, Stop, Play, Update, Install, Cancel, Details, Settings, Logs, Edit Game, Hide or Unhide, Favourites, Categories, Remove From Recent, Uninstall.

Edit Game (v2.22): rename any game and set its card and cover images, with SteamGridDB picking when an API key is set.

Game page and actions

The game page is a full-bleed background with a cover column, an info column with playtime, one main button and four tabs; most management actions live in a ⋮ menu.

Layout

- Background from `art_background`, falling back to `art_cover`; top row with back, settings and ⋮ menu.
- Left: 3:4 `art_square` (600×800) with logo overlay.
- Info: store icon, genres, developer, release date, description; Last Played, Time Played, First Played; status line, launch-option selector and main button.
- Main button states: Install, Import Game, Play, Play with logs, Stop, Cancel, Remove from Queue, plus Installing Redistributables, Installing Winetricks Packages, Syncing saves.

Tabs

Tab	Contents
Install info	Download and install size, platform, version, online required, third-party manager, path, cloud saves (sync state, open saves folder)
Achievements	GOG only (since v2.21)
Extra info	Scores, HowLongToBeat, ProtonDB tier and Steam Deck compatibility, AppleGamingWiki
Requirements	System requirements

An anticheat info block and a Report Issue link sit on the page too.

⋮ **menu**: Edit Game, add or remove shortcut, Uninstall, Force Update, Move Game, Change Install Location, Verify and Repair, EOS Overlay, Add to or Remove from Steam, Categories, Store page, Changelog, Check Compatibility, Requirements, Modify Installation (DLC, language, GOG branch or build pinning), Browse Files, Browse Wine Prefix.

Install dialog: platform, Wine version, prefix (suggested `<prefix dir>/<Title>`), DLC, language; an Import button beside Install.

Metadata sources: PCGamingWiki, GamesDB, AppleGamingWiki, ProtonDB summaries by Steam app ID, the Steam Deck compatibility endpoint, HowLongToBeat, the umu API (for GAMEID); SteamGridDB for art.

Logs per game and type (launch, install, import, repair, update, setup); "Play with logs" opens them and logs can be uploaded to dpaste.

Settings model

Heroic stores global defaults in `config.json` and per-game overrides as diffs in `GamesConfig/<appName>.json`; the per-game modal has six tabs, which is more than our app should copy.

Global (`config.json` → `defaultSettings`)

- General: language, theme, default install path (`~/Games/Heroic`), prefix base (`~/Games/Heroic/Prefixes`), Steam path, EGS sync, update checks, auto-update games, start in Console Mode, tray, minimize on launch, shortcuts (desktop, start menu, Steam), playtime sync, Discord RPC, disable controller, library top section, max recent games.
- Advanced: SteamGridDB API key, alternative backend binaries, gamepad repeat delays, hide window on `heroic://` launch, experimental features, custom CSS, clear cache, reset.
- "Game Defaults" reuses the per-game panel but only applies at install time; installed games must be changed from their own settings.

Per-game file: `{ "<appName>": {overrides}, "version": "v0", "explicit": false }`; values equal to the global default are dropped, so only differences are stored.

Tab	Settings
Wine	Wine/Proton version (`wineVersion` with bin, name, type wine, proton, crossover or toolkit), prefix, auto-install DXVK, DXVK-NVAPI, VKD3D; Esync, Fsync, Wine-Wayland, WoW64, FSR; winecfg, Winetricks, run EXE in prefix
Other	DXVK FPS HUD, MangoHud, GameMode (default on in Flatpak), prefer system libs, Steam Runtime (scout, soldier, sniper, steamrt4), NVIDIA Prime, BattlEye and EAC runtimes
Advanced	Ignore updates, offline mode, verbose logs, alternative EXE, launch option, launcher args, before and after scripts, wrappers table, environment variables table, preferred language
Cloud Saves Sync	Auto sync, save paths, manual download, upload, force download, force upload
Gamescope (Linux)	Upscaling (width, height, method), window type, FPS limiter focused and unfocused, force grab cursor, extra options
Legacy	DXVK FPS limit, Disable UMU (mostly deprecated)

Wine Manager: Linux tabs for Proton-CachyOS and GE-Proton (Proton-CachyOS downloaded on first run); search, release date, size, install and remove. Discovery also scans `~/.config/heroic/tools/{wine,proton}`, every Steam library's `compatibilitytools.d`, Lutris Wine runners and custom paths.

umu: used when the Wine type is Proton and umu is not disabled; Heroic prefers `umu-run` on PATH and otherwise downloads it through the Lutris runtimes API.

Wrapper variables passed to launches: `HEROIC_APP_NAME`, `HEROIC_APP_RUNNER`, `HEROIC_APP_SOURCE`, `STORE` (egs, gog, amazon, zoomplatform), `GAMEID=umu-0`.

Add to Steam writes Steam's `shortcuts.vdf` with grid art, optionally on every install.

Controller support

Heroic has two controller paths: spatial navigation over the normal UI, and since v2.21 a dedicated fullscreen Console Mode that is closer to what our couch mode needs.

Normal UI (`helpers/gamepad.ts`)

- Gamepad API spatial navigation with a hints bar and an on-screen keyboard (simple-keyboard) that opens on text inputs.
- A is the main action, B goes back, Y on a card plays or installs, the right stick scrolls, and Guide toggles Console Mode.
- Settings: disable controller, repeat delay 50 ms, initial repeat delay 300 ms.

Console Mode (route `/console`)

- Fullscreen, no sidebar; entered from the sidebar, the Guide button, a start-up setting or the `--console` flag.
- Grid of 3:4 `art_square` cards, installed and not, with in-card progress bars; hidden games and DLC excluded.
- L1/R1 cycle the store filter (All, Epic, GOG, Amazon, Other, ZOOM); R2 toggles A–Z or Z–A; an installed-only filter exists.
- Top bar: Sort, Quit Console, Quit App.
- Launch overlay steps through syncing saves, applying Winetricks, launching, playing; holding a button for 3 s cancels.
- Install overlay (platform, Wine, path), update notice and cancel-download dialogs.
- Detects PlayStation, Xbox, Nintendo and Steam Deck layouts; swaps A and B on Nintendo controllers.

Steam Deck Game Mode is detected through `XDG_CURRENT_DESKTOP=gamescope`; Heroic then skips its own MangoHud and Gamescope.

Accessibility page: zoom, content and action font families, all tiles in colour, always show titles, disable click-outside closing, disable smooth scrolling, disable animations, 18 themes including high contrast. No screen-reader settings beyond Electron's basic ARIA.

Data files and launch integration

Everything an importer needs is electron-store JSON under one root: `~/.config/heroic/` natively, `~/.var/app/com.heroicgameslauncher.hgl/config/heroic/` for the Flatpak (confirmed in Heroic's `paths.ts` and Cartridges' candidate list).

Path (under root)	Contents
`config.json`	`{defaultSettings, version}`
`GamesConfig/<appName>.json`	Per-game overrides
`store/config.json`	`games.recent`, `games.hidden`, `games.favourites` as `[{appName, title}]`; `games.customCategories` as `{name: ["<appName>_<runner>"]}`; theme, window state, user info
`store/timestamp.json`	`{<appName>: {firstPlayed, lastPlayed (ISO), totalPlayed (minutes)}}`
`store/game-overrides.json`	User-edited `title`, `art_cover`, `art_square` (v2.22)
`store_cache/legendary_library.json`	`{library: GameInfo[]}` (Epic)
`store_cache/gog_library.json`	`{games: GameInfo[]}`
`store_cache/nile_library.json`	`{library: GameInfo[]}` (Amazon)
`store_cache/zoom_library.json`	`{games: GameInfo[]}`
`sideload_apps/library.json`	`{games: GameInfo[]}`
`legendaryConfig/legendary/installed.json`	Installed Epic games keyed by app name

Path (under root)	Contents
`gog_store/installed.json`	`{installed: [{appName, install_path, platform, version}]}` (`gog_store/auth.json` and `config.json` hold credentials: never read)
`nile_config/nile/installed.json`	Installed Amazon games
`zoom_store/installed.json`	Installed ZOOM games
`images-cache/<sha256(url)>`	Cached art; Epic URLs get `?h=400&resize=1&w=300` appended before hashing

Useful `GameInfo` fields: `runner`, `app_name`, `title`, `art_cover` (landscape), `art_square` (portrait), `art_logo`, `art_background`, `art_icon`, `developer`, `is_installed`, `install{install_path, platform, install_size, version}`, `canRunOffline`, `is_linux_native`, `extra{about, genres, releaseDate}`, `cloud_save_enabled`.

Launching a Heroic game from outside

- URI: `heroic://launch?appName=<id>&runner=<legendary|gog|nile|sideload>[&arg=…][&altExe=<path>][&gui=false]`; legacy `heroic://launch/<runner>/<appName>`; `heroic://ping` for testing. Not-installed games prompt to install.
- The URI runner list does not include `zoom`, so ZOOM games apparently cannot be launched by URI (code observation).
- CLI flags: `--no-gui`, `--console`, `--fullscreen`.
- Heroic's own Steam shortcut uses `flatpak run com.heroicgameslauncher.hgl --no-gui --no-sandbox "heroic://launch?appName=X&runner=Y"`; its desktop shortcuts use `xdg-open heroic://launch?…`.

Other locations: logs in `$XDG_STATE_HOME/Heroic/logs/games/<appName>_<runner>/<type>.log`; prefixes in `~/Games/Heroic/Prefixes/`.

Prior art: Cartridges' `heroic_source.py` reads sideload, Epic, GOG and Amazon (not ZOOM), honours Heroic's hidden list, imports only installed games, takes covers from `images-cache` and launches with the legacy URI. Lutris has no Heroic import; it only bundles gogdl for GOG downloads.

Flatpak packaging and permissions

Heroic runs Wine and Proton inside its own sandbox (no host spawning), which costs it `--device=all`, `--allow=devel` and multiarch. Manifest: flathub/com.heroicgameslauncher.hgl.

Item	Value
Runtime	`org.freedesktop.Platform` 26.08 with `org.electronjs.Electron2.BaseApp`
Entry	`heroic-run`: links Discord IPC sockets, sets TMPDIR, runs `zypak-wrapper`; app module is the extracted release AppImage
Extra modules	vulkan-tools, rsync, 7zip, cabextract, unrar (Winetricks), TiMidity++ with a soundfont; i386 Compat, GL32, VAAPI i386 extensions
Allow, devices	`devel`, `multiarch`, `bluetooth`, `per-app-dev-shm`; `--device=all`
Sockets	network, ipc, x11, wayland, pulseaudio
Filesystem	`~/Games/Heroic:create`, `~/.steam`, `~/.var/app/com.valvesoftware.Steam`, `~/.local/share/lutris`, `~/.local/share/applications` (read-write); documents, desktop, `/run/media`, `/mnt`, `/media`; `/run/udev:ro`; `xdg-run/gamescope-0`; `xdg-data/umu:create`; `xdg-config/MangoHud:ro`
D-Bus	UDisks2, UPower, Notifications, ScreenSaver, StatusNotifierWatcher, MediaKeys

Item	Value
Environment	`PATH` includes the MangoHud, gamescope and OBS Vulkan capture extension dirs, so those must be installed as Flatpak Vulkan-layer extensions matching 26.08

umu and pressure-vessel work inside the sandbox through `devel`, `multiarch` and `per-app-dev-shm`, with umu data shared at `xdg-data/umu`.

Adopt, adapt, avoid

Read Heroic's files and hand launches back to it; take its card states and Console Mode ideas, and leave its settings sprawl behind.

Adopt

- Import from `store_cache/*_library.json` joined with each store's `installed.json`, then apply `store/game-overrides.json` so user-edited titles and art carry over.
- Read `store/timestamp.json` for playtime and last played, and `store/config.json` for hidden, favourites and categories.
- Launch with `heroic://launch?appName=…&runner=…&gui=false`, or `flatpak run com.heroicgameslauncher.hgl --no-gui "heroic://…"` on the host.
- Portrait 3:4 `art_square` for the grid, landscape `art_cover` or `art_background` for hero and continue-playing tiles; this matches an Xbox-style hero lane.
- Status badges on cards (install progress, update available, playing) and a hold-to-cancel launch overlay.
- From Console Mode: controller glyph detection (Xbox, PlayStation, Nintendo with A/B swap, Deck), L1/R1 to switch filters, a bottom hint bar.

Adapt

- Replace the seven-toggle filter dropdown with a chip row or toggle group (Installed, Favourites, a store popover) plus a sort menu, and add the sorts Heroic lacks: last played, playtime, recently added.
- Recreate the top lane (recently played, favourites) as a horizontal carousel.

- Turn the tabbed game page into a navigation page: hero banner, primary Play or Install button, then preference groups for info, compatibility (ProtonDB, Deck, HowLongToBeat) and saves.
- Show Heroic's per-game settings read-only and open Heroic to edit them, rather than re-implementing them.

Avoid

- Six per-game settings tabs plus a Legacy tab, and "Game Defaults" that only apply at install.
- Hover-only actions and grayscale for not-installed games (Heroic itself added an opt-out).
- A sidebar full of donation and documentation links.
- Writing to Heroic's files, reading its credential files, or relying on ZOOM URI launches.

Open questions

- Will the OpenURI portal open `heroic://` from a sandboxed app without the Flatpak talk-name?
- Are `game-overrides.json` keys bare app names or `appName_runner`?
- Where exactly do Heroic Flatpak logs live?

Sources

- HeroicGamesLauncher repository (README, package.json)
- Backend: protocol.ts, constants/paths.ts, constants/environment.ts, storeManagers, cache.ts, config.ts, game_config.ts, launcher.ts, compatibility_layers.ts, logger/paths.ts, shortcuts, images_cache.ts
- Common types: types.ts, electron_store.ts
- Frontend: screens, helpers/gamepad.ts
- Packaging: Flatpak manifest template, electron-builder.yml, Flathub manifest repo, Flathub app page
- Releases: all, v2.22.4, v2.22.0, v2.21.0
- Heroic wiki and Gamepad Navigation page (outdated)
- Heroic official site
- GamingOnLinux: Heroic v2.19 released
- Cartridges heroic_source.py

- Lutris gogdl bridge